

❖ FIFA World Cup 2026 - Finalist Prediction

Week 1: Data Collection & Cleaning

1. Data source:

- **Kaggle** - 900+ World Cup matches (1930-2022)
- [FBref.com](https://fbref.com) - Team stats (possession, shots, goals)
- [FIFA.com](https://fifa.com) - Official team rankings
- **ESPN** - Match statistics

Total: 640+ teams across 20+ tournaments

2. Web Scraping

Built a Python script to automatically extract data from [FBref.com](https://fbref.com):

- Extracted match dates, teams, goals, possession %
- Got player ages and international experience
- Saved everything as CSV files
- Used: requests, BeautifulSoup, pandas

3. Data Cleaning

Fixed the messy data:

- Removed 32 duplicate entries
- Filled missing ages with averages
- Fixed date formats
- Removed incomplete records

4. Features Created

Made 7 new important features:

Feature	Means
FIFA Ranking	Official strength score
Avg Age	How young/old the team is
Goal Difference	Goals scored - conceded
Win Rate	% of games won
Player Experience	Total player international matches
Recent Form	Performance last 10 games
Home Advantage	Hosting tournament (1=yes, 0=no)

5. Final Dataset Size:

640 rows × 15 columns

Sample:

Team	Year	Rank	Age	Goal_Diff	Win%	Experience	Finalist?
Brazil	2022	1	28.5	+12	89%	450	YES
Argentina	2022	3	26.2	+8	85%	380	YES
France	2022	4	27.1	+5	78%	420	NO
Germany	2022	11	27.1	+2	71%	520	NO

6. Problems & Solutions

Problem	Fix
Website blocking scraper	Added request delays
Team name inconsistency	Used FIFA codes only
Missing player ages	Used team average
Old data incomplete	Used only complete records